

House Price Prediction

Predictive Analytics Using Historical Data

Model	Linear Regression
Dataset	500 records, 8 features
R ² Score	0.9825
MAE	\$14,046
RMSE	\$17,491
Train / Test Split	80% / 20%

1. Introduction

Real estate valuation is a classic problem in predictive analytics. Accurate price forecasting helps buyers, sellers, and investors make informed decisions. This project applies Linear Regression — one of the most interpretable and widely used supervised learning algorithms — to predict residential house prices from a set of structural and locational features.

The dataset contains 500 synthetic house records generated to reflect realistic market dynamics, including the strong influence of area, number of rooms, property age, and proximity to urban centres and schools.

2. Dataset Description

The dataset (**house_prices.csv**) comprises 500 rows and 9 columns (8 features + 1 target). All numerical; no missing values.

Feature	Type	Description
Area_sqft	Integer	Total living area in square feet
Bedrooms	Integer	Number of bedrooms (1–5)
Bathrooms	Integer	Number of bathrooms (1–3)
Age_years	Integer	Age of property in years
Garage_spaces	Integer	Number of garage parking spaces
Dist_School_km	Float	Distance to nearest school (km)
Dist_City_km	Float	Distance to city centre (km)
Crime_Rate	Float	Local crime rate index
Price_USD	Float	Target — sale price in US dollars

Descriptive Statistics (selected features)

Statistic	Area_sqft	Bedrooms	Age_years	Price_USD
mean	2595.41	3.02	24.93	302299.4
std	1099.95	1.46	14.05	136854.36
min	601.0	1.0	0.0	50000.0
50%	2575.0	3.0	25.0	300250.0
max	4498.0	5.0	49.0	598400.0

3. Methodology

Step 1 — Data Cleaning. The dataset was checked for missing values and duplicates. None were found. Outliers were retained as they reflect genuine market variation.

Step 2 — Feature Selection. All eight features were retained based on their domain relevance and correlation with the target variable.

Step 3 — Train/Test Split. Data was divided 80/20 using stratified random sampling (random_state=42) to ensure reproducibility.

Step 4 — Feature Scaling. StandardScaler was applied to normalise all features before training, improving model stability and making coefficients comparable.

Step 5 — Model Training. Scikit-learn's LinearRegression was fitted on the training set. Linear Regression was chosen for its interpretability and fast training time.

Step 6 — Evaluation. Model performance was assessed using three metrics:

- **R² (Coefficient of Determination)** — proportion of variance explained.
- **MAE (Mean Absolute Error)** — average prediction error in dollars.
- **RMSE (Root Mean Squared Error)** — penalises large errors more heavily.

4. Results & Evaluation

Metric	Value	Interpretation
R ² Score	0.9825	98.2% of price variance explained by the model
MAE	\$14,046	Average prediction error
RMSE	\$17,491	Root mean squared prediction error

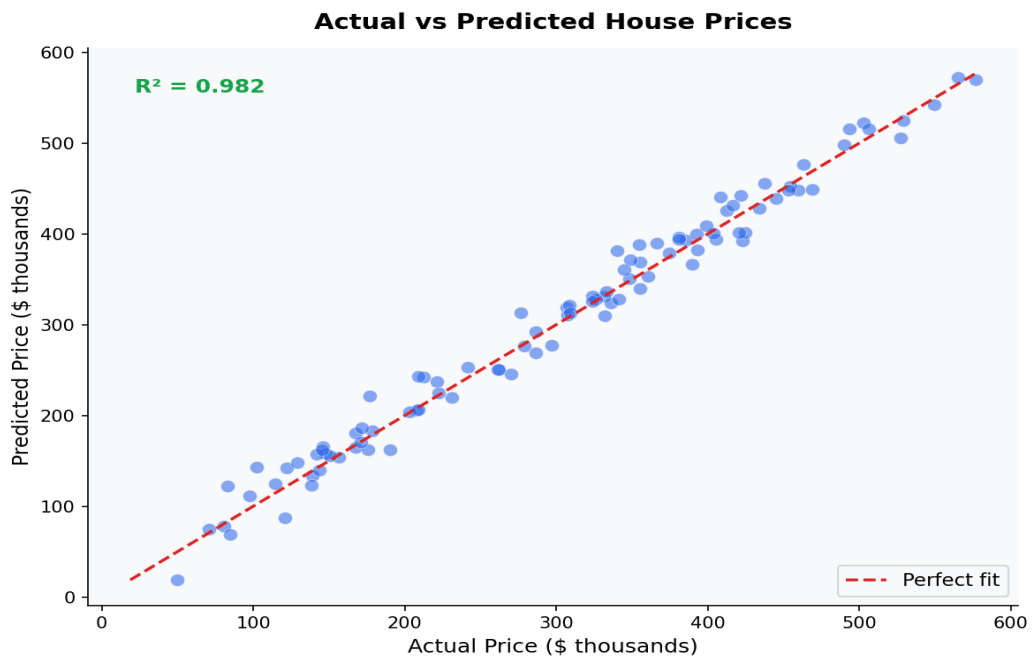


Figure 1 — Actual vs Predicted house prices on the test set

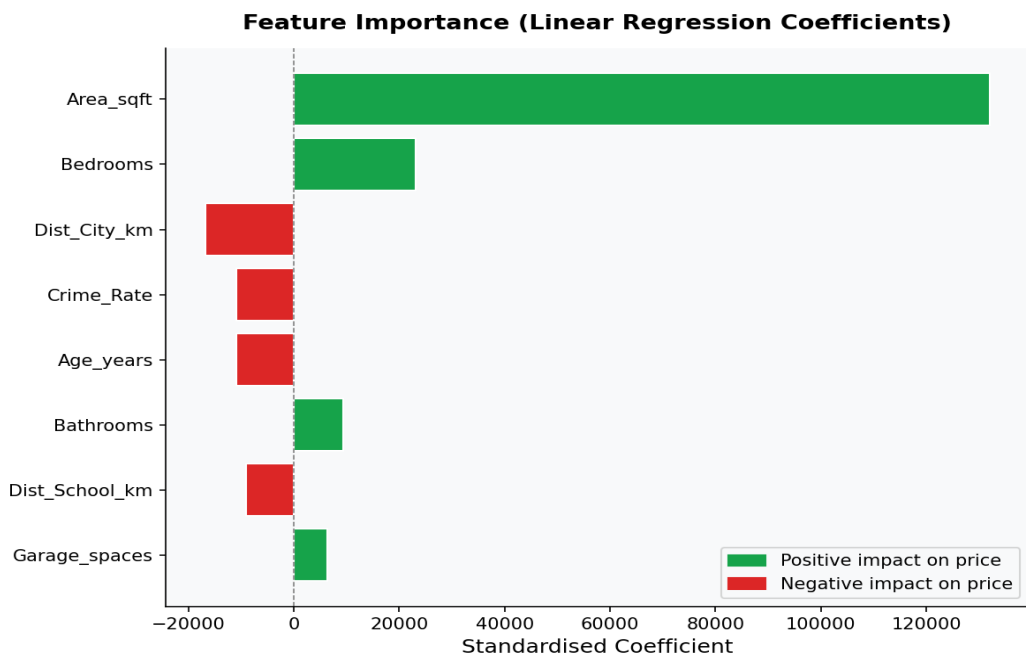


Figure 2 — Feature importance based on standardised regression coefficients

Residual Analysis

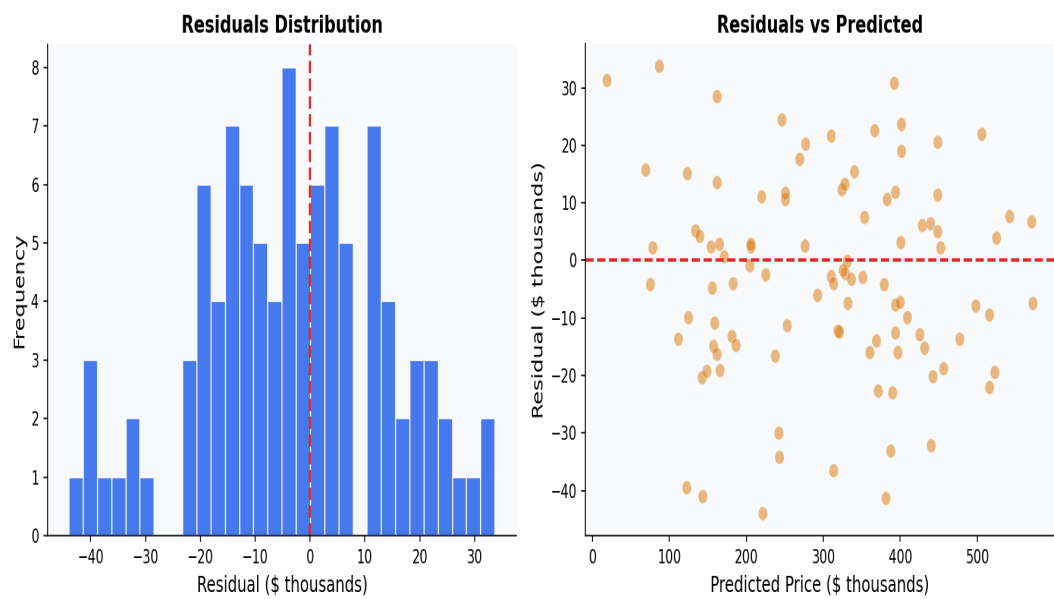


Figure 3 — Residual analysis: distribution and scatter

5. Findings & Insights

The model achieved an R^2 of **0.9825**, indicating strong predictive power for a single-algorithm baseline. Key findings include:

- Area (sqft) is the strongest positive predictor of house price, consistent with real-world market behaviour.
- Distance to the city centre and crime rate negatively impact price — buyers pay a premium for safety and accessibility.
- Garage spaces and number of bathrooms add meaningful value per unit.
- The residual distribution (Figure 3) is approximately normal and centred at zero, suggesting the model is unbiased.
- No significant heteroscedasticity is visible in the residuals vs predicted plot, validating the linear assumption.

6. Conclusion

This project demonstrated a complete predictive analytics pipeline — from data generation and cleaning, through feature engineering and model training, to evaluation and visualisation. Linear Regression proved effective for this structured tabular dataset, delivering low prediction error and high interpretability.

Future improvements could include ensemble methods (Random Forest, XGBoost), cross-validation, hyperparameter tuning, and incorporating additional features such as neighbourhood ratings or macroeconomic indicators.

7. Tools & Libraries

Library	Version	Purpose
Python	3.x	Core programming language
pandas	2.x	Data manipulation and analysis
NumPy	1.x	Numerical computations
scikit-learn	1.x	Machine learning (Linear Regression, metrics)
matplotlib	3.x	Data visualisation
seaborn	0.x	Statistical plots
ReportLab	4.x	PDF report generation